

§2.2 Energy Model

Prepared milk is used as the direct proxy for caloric energy: 1 ml of prepared milk = 1 unit of energy. The baby's daily nutritional need and the resulting hourly drain rate are:

$$D = w \cdot r \quad \lambda = D/24$$

w = baby weight (kg), $r=150$ ml/kg/day (default), λ = hourly drain rate (ml/h). Example:
 $w=6.9$ kg $\rightarrow D=1035$ ml, $\lambda=43.1$ ml/h.

If the parent feeds one standard bottle of m_0 ml of prepared milk every SI hours, intake stays in equilibrium. The standard interval is the time for that bottle to be fully spent at rate λ :

$$SI = m_0 / \lambda$$

90 ml water $\rightarrow m_0=100$ ml milk, $\lambda=43.1$ ml/h $\rightarrow SI \approx 2\text{h } 19\text{m}$

§4.1 The Intake Function

Rather than tracking an all-time energy balance, the app tracks the total milk received in a rolling 24-hour window. The credit contributed by bottle i (volume m_i , given at time t_i) evaluated at time T is:

$$c_i(T) = m_i \text{ if } T - t_i \leq 24 \text{ h} \quad \max(0, m_i - \lambda(T - t_i - 24) \text{ if } T - t_i > 24 \text{ h}$$

The smoothed 24-hour intake at time T is then the sum over all feeds logged before T :

$$I(T) = \sum_{i=1}^n c_i(T)$$

The decay rate past 24 h is exactly λ — the same rate at which the baby burns energy. This is not arbitrary: a bottle's contribution to the window diminishes at the same speed as its nutritional value is consumed. In steady-state feeding, $I(T)$ oscillates between D (just before each feed) and $D+m_0$ (just after).

§5.1 Predictor A — Adjusted Volume at Standard Time

Predictor A answers: if I feed at the standard time, how much should I give? It computes the volume that restores intake to the equilibrium peak $D+m_0$:

$$TA = t_{last} + SI$$

$$VA = D + m_0 - I(TA)$$

At equilibrium $I(TA)=D$, so $VA=m_0$ — the standard bottle.

The parent is shown VA converted to water ml and decides which bottle size to use. If $VA \leq 0$ the baby is still overfed at the standard time — Predictor A is greyed out. If VA exceeds the stomach capacity cap it is clamped (see §6).

§5.2 Predictor B — Standard Volume at Adjusted Time

Predictor B answers: when should I give my preferred standard bottle? It finds the time TB at which $I(T)$ has decayed back to D , so that adding m_0 restores intake to $D + m_0$:

$$I(TB) = D$$

Feed when the 24-hour intake returns to the daily target.

TB is found by binary search on $[t_{last}, t_{last} + T_{max}]$. At equilibrium $TB = t_{last} + SI$. With a deficit, $TB < t_{last} + SI$; with a surplus, $TB > t_{last} + SI$.

§6 Stomach Capacity Constraint

The total undigested milk in the stomach at time T is modelled as the sum of unmetabolised portions of all recent feeds:

$$S(T) = \sum_{i=1}^n \max(0, m_i - \lambda(T - t_i))$$

The constraint is $S(TB) + m_0 \leq m_{cap}$, where m_{cap} is one standard bottle size above the preferred size. Assuming only the most recent feed is still undigested, this gives a minimum waiting time before the next feed:

$$\Delta t_{min} = \max(0, m_{last} + m_0 - m_{cap}) / \lambda$$

Example: $m_{last} = m_0 = 100$ ml, $m_{cap} = 135$ ml, $\lambda = 43.1$ ml/h $\rightarrow \Delta t_{min} = (100 + 100 - 135) / 43.1 \approx 1$ h 30m

If Predictor B's TB falls before $t_{last} + \Delta t_{min}$, it is clamped upward with a note: "Stomach not ready — baby is underfed but needs time to digest."